Clustering Strategy for Megaconstellation With Synergistic Energy and Size Considerations

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This article proposes an energy- and size-aware clustering (ESC) strategy for megaconstellation systems. The methodology constructs a hierarchical architecture with designated cluster heads and member satellites through role differentiation. Cluster heads serve as relay nodes, reducing the communication load on ground infrastructure by interacting with ground stations and other entities. This hierarchical approach enables efficient constellation control through interactions primarily with cluster heads. The ESC strategy begins with a cluster head election process, incorporating orphan handling management to select cluster heads, ensuring full coverage of the entire constellation. Then, a two-stage multiobjective optimization framework is used to deal with the division of members into clusters and intracluster multihop connectivity configurations. This dual-phase approach optimizes energy consumption while maintaining balanced cluster sizes. Simulations on the OneWeb system validate the ESC strategy as a viable solution in that it achieves 55.9% energy consumption reduction compared to traditional discrete gradient optimization methods while maintaining more balanced cluster sizes.

I. INTRODUCTION

A. Motivation

Megaconstellation systems, such as Starlink, OneWeb, and Telesat, have attracted significant interest in the aerospace community [1], [2], [3]. These vast networks, composed of thousands of low-orbiting satellites, offer cost-effective components, reduced latency, and comprehensive coverage [4], [5]. Despite their advantages, megaconstellations present significant management challenges due to their rapid movement and time-varying topology [6], [7].

Traditional constellations (e.g., GPS, GLONASS, and Galileo) with limited satellites can be centrally managed by a few ground stations. In contrast, megaconstellations with thousands of satellites require a more complex management system beyond single ground station control. To facilitate efficient management, megaconstellations are often clustered using clustering strategies [8]. As shown in Fig. 1, for the OneWeb system using the proposed energy- and size-aware clustering (ESC) strategy in this article, satellites are categorized as cluster heads or members based on functionality. Cluster heads manage member satellites and communicate with ground stations, while members communicate only with their respective heads. This clustered approach reduces the workload on ground stations by enabling them to manage member satellites indirectly via cluster heads.

B. Related Works

Cluster-based management has been proven effective in managing large network systems. This paradigm is extensively investigated in wireless sensor networks and mobile ad hoc networks [9], [10], [11], indicating its critical significance in improving network performance and efficiency [12], [13]. Heinzelman et al. [14] partitioned the network into multiple clusters using a stochastic cluster head selection process based on the low-energy adaptive clustering hierarchy (LEACH) algorithm [14]. Wei et al. [15] proposed an energy-efficient clustering method

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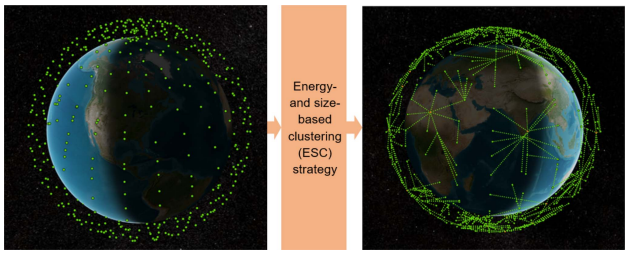


Fig. 1. Scenarios for clustering cases on OneWeb constellations based on the proposed ESC strategy.

and determined the appropriate cluster based on the hop distance from the base station. Panchal and Singh [11] introduced the energy-efficient hybrid clustering and hierarchical routing algorithms for wireless sensor networks. The algorithm significantly enhances network lifetime and coverage while optimizing energy consumption. Panchal and Singh [13] proposed an energy-efficient technique for selecting optimal cluster heads and grid heads. It enhances network lifetime, coverage, and energy efficiency through dynamic grid head selection and a fitness function based on residual energy, distances, and grid-centroid locations.

Compared to the static framework of wireless sensor networks, the dynamic nature of nodes in mobile ad hoc networks introduces complexity to the clustering methodology, thus posing more challenges [16]. Zhao et al. [17] proposed a fuzzy C-means algorithm to select cluster heads based on vehicle movement direction, weighted mobility, and entropy. Chatterjee et al. [18] proposed a weighted clustering algorithm (WCA) that selects cluster heads by evaluating node connection, mobility, transmission power, and battery power. In addition, intelligent swarm optimization algorithms are often employed in cluster strategy. For example, in [19] and [20], the ant colony optimization algorithm and the modified particle swarm algorithm are used to select cluster heads. Although clustering strategies are well utilized in large-scale networks, their application to megaconstellations introduces unique complexities that demand further investigation. First, megaconstellation networks exhibit much faster topology changes than wireless sensor networks or mobile ad hoc networks due to the low orbital altitude of their satellites, resulting in higher relative motion speeds [6], [9], [17]. Next, during orbital operations, satellites face battery capacity limitations, and the long distances between them lead to higher communication energy consumption compared to sensor networks. As a result, clustering strategies must be carefully designed to minimize energy consumption [21]. In addition, traditional wireless sensor networks typically consider only distance constraints for communication between nodes [13]. However, in constellation scenarios, satellite communication must account for both distance constraints and ground obstacles, making the clustering process more complex. Finally, when designing the clustering strategy for constellations, rationalizing the number of clusters is crucial. This process must balance the operational burden on ground management with the overall network management efficiency.

Megaconstellation clustering remains largely unexplored. Ma et al. [22] proposed a clustering strategy based on a multilayer network structure. The strategy uses medium-orbit and cluster-head satellites to control the massive constellation network together. Li et al. [8] proposed a distributed domain control strategy using a multifactor weighted approach and integrating the satellite's speed, energy, and neighborhood degree to select head satellites. Using coalition game theory, Liu et al. [23] selected cluster-head satellites automatically and totally distributedly. In [8] and [23], the number of established clusters accounts for 10%–15% of the satellite network's nodes. Managing 1200–1800 clusters for megaconstellations, such as Starlink, which aims to deploy 12 000 satellites [24], [25], presents a substantial challenge for ground-based control facilities.

Existing clustering strategies often neglect the dual objectives of minimizing cluster count and optimizing intracluster energy. This article prioritizes neighborhood degree and energy (NE) to divide satellites into clusters while reducing cluster count and intracluster energy consumption.

C. Contribution

1) To reduce the operational load on ground control facilities and improve management efficacy, we propose the ESC strategy, comprising an NE-based cluster head election method augmented with an orphan node handling procedure. Compared to traditional methods [22] and [23], our strategy results in a reduced cluster quantity.
2) Compared to traditional methods [14], [22], where constellation clustering solely considers cluster size, the ESC strategy also contains a multiobjective framework for dividing members into clusters and establishing intracluster multihop connections.
3) We propose a cluster size index in the ESC framework to quantify cluster balancing. Compared to the general load-balancing index [26], which only considers the degree of cluster balancing, the proposed index further considers the cluster size.

II. ACCESS- AND ENERGY-BASED MODELING

This section discusses the natural cluster structure of megaconstellations, analyzes the access model within clusters, and then delves into the energy consumption model, a critical aspect of constellation management.

A. Clustering in Constellation

The management structure of the megaconstellation is depicted in Fig. 2.

Given the large number of satellites forming the megaconstellation—numbering in the thousands—the traditional approach to constellation management, which mostly depends on single-point control of satellites via

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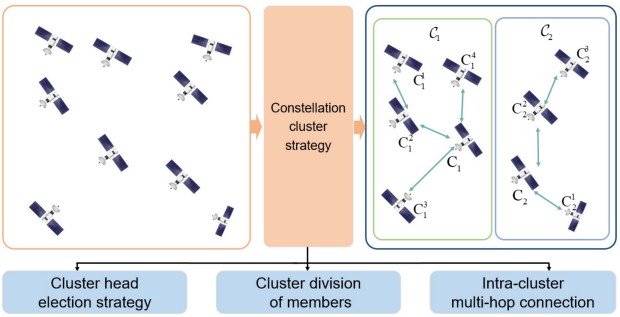


Fig. 2. Megaconstellation management structure.

ground stations, appears impractical. To handle this complexity, we used a clustering approach that divided the megaconstellation into manageable clusters. This approach consists of the following three steps.

1) Elect cluster heads using a specialized method.
2) Optimization of cluster division for member satellites.
3) Creating intracluster multihop links to enable efficient communication.

Using a head election strategy, we select H cluster heads in a constellation containing N satellites. Define cluster head set $C = \{C_1, \ldots, C_i, \ldots, C_H\}$, where C_i is the corresponding head of cluster-i. The other satellites act as cluster members. Define cluster member set $\mathcal{M} = \{\mathcal{M}_1, \ldots, \mathcal{M}_m\}$.

We partition the members into clusters, ensuring that each cluster comprises one head and multiple members. After dividing the members into corresponding clusters, define the set of clusters $\mathcal{C} = \{\mathcal{C}_1, \ldots, \mathcal{C}_i, \ldots, \mathcal{C}_H\}$. For a given $\mathcal{C}_i$, members C_i^j were incorporated into a designated cluster set, where j signifies the individual members in $\mathcal{C}_i$. For cluster $\mathcal{C}_i$ with J members, $\mathcal{C}_i = C_i \cup (\cup_{j=1}^{J} C_i^j)$, i.e., cluster $\mathcal{C}_i$ includes a head C_i and J members C_i^j. In a cluster, communication between members and the head can be achieved via single-hop or multihop connections.

1) *Single-Hop Connections:* Members are typically directly linked to the heads when they are close to the heads. As shown in Fig. 2, C_1^3 is directly connected to C_1.
2) *Multihop Connections:* Members can use other members in the cluster as relays to connect to the cluster head when they are far away. For example, in Fig. 2, C_1^2 serves as a relay between C_1 and C_1^1. A satellite that requires a multihop connection is defined as a *source satellite*, such as C_1^1 in Fig. 2, while satellites providing relay services are defined as *relay satellites*, such as C_1^2 in Fig. 2.

Each satellite has a unique ID. The uniqueness ID guarantees that identity is unambiguous even when clusters are realigned within the constellation.

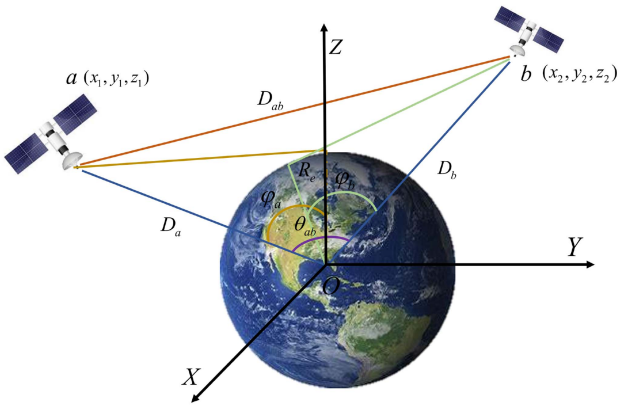


Fig. 3. Geometric visibility between satellites.

B. Access-Based System Model

Assuming quasi-stationary satellite positions and velocities over a short time period T, we simplify constellation topology analysis by treating position and velocity as constant within T. Mutual visibility is essential for establishing access and intracluster links between satellites, making network construction critical for modeling the constellation's structure. At the start of T, we compute visibility using satellite positional coordinates. Fig. 3 shows the geometric line-of-sight relationship between satellites. The positions of satellites a and b are (x_1, y_1, z_1) and (x_2, y_2, z_2). The distance D_{ab} is

$$D_{ab} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2} \tag{1}$$

and $D_a = \sqrt{x_1^2 + y_1^2 + z_1^2}$ and $D_b = \sqrt{x_2^2 + y_2^2 + z_2^2}$ are the orbital radius w.r.t. the Earth center O. $D_{\max}$ is a satellite's permitted maximum communication range. The angle $\theta_{ab} = \angle aOb$ is

$$\theta_{ab} = \cos^{-1}\left(\frac{x_1x_2 + y_1y_2 + z_1z_2}{D_aD_b}\right). \tag{2}$$

The angles φ_a and φ_b of a and b to the Earth are

$$\varphi_a = \cos^{-1}(R_e/D_a), \quad \varphi_b = \cos^{-1}(R_e/D_b) \tag{3}$$

where R_e is Earth radius. Satellites a and b are mutually visible when $\varphi_{ab} = \varphi_a + \varphi_b > \theta_{ab}$ and $D_{ab} < D_{\max}$ are satisfied.

The graph theory is used to describe the constellation topology. The N-satellite constellation is modeled as a graph $G(\mathcal{N}, \mathcal{E}, A)$, where $\mathcal{N} = \{n_1, \ldots, n_N\}$ is the set of satellite nodes, $\mathcal{E} = \{e_{ij} \mid i \in \mathcal{N}, j \in \mathcal{N}, i \neq j\}$ is the edges set, and $A = \{0, 1\}^{|\mathcal{N}|\times|\mathcal{N}|}$ is the adjacency matrix [27]

$$A = \begin{bmatrix} 0 & A_{12} & \ldots & A_{1N} \\ A_{21} & 0 & \ldots & A_{2N} \\ \vdots & \vdots & & \vdots \\ A_{N1} & A_{N2} & \ldots & 0 \end{bmatrix} \tag{4}$$

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where A_{ab} ($a \in \mathcal{N}$ and $b \in \mathcal{N}$) is a Boolean variable, denoting the edge weight between the nodes a and b

$$A_{ab} = \begin{cases} 1, & \text{if } (n_a, n_b) \in \mathcal{E} \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

C. Energy Consumption Model

At the start of each T, satellites in the network exchange information via radio waves. For cluster $\mathcal{C}_i$, cluster head C_i broadcasts a *hello* message to each member in $\mathcal{C}_i$, including its geolocation and ID. Upon reception, each member responds with an equivalent *hello*, informing the head as well. Radio communications between satellites are the main source of energy consumption in the constellation [8], [21]. The energy required to transmit l-bits data is modeled by

$$\begin{cases} E_{\text{tx}} = lE_{ele} + l\varepsilon_{\text{amp}}D^2 \\ E_{\text{rx}} = lE_{ele} \end{cases} \tag{6}$$

where E_{tx} and E_{rx} denote the energy consumption for transmission and reception, E_{ele} is the energy consumption of the transmitter and receiver modules in transmitting and receiving 1-bit data, ε_{amp} denotes the radio-frequency amplifier gain, and D denotes the transmitting distance. We assume that each satellite carries equal initial energy $E_{\max}$.

REMARK 1 Most satellites can replenish their battery energy through solar panels during operation. However, when applying the ESC strategy for clustering, battery charge is treated only as an input parameter. In practice, whether the battery is charging or discharging does not affect the algorithm, and the charging process does not impact satellite operations. The system only needs to provide the current battery level as input. For convenience in clustering calculations, we assume a total energy value, but this does not affect the algorithm's usability or performance.

D. Hybrid Discrete Multiobjective Particle Swarm Optimization (HD-MOPSO) Algorithm

PSO, a meta-heuristic global optimization method based on population intelligence [28], offers several advantages over other meta-heuristic optimization methods. Its benefits include rapid convergence, broad applicability, simplicity, and ease of implementation. PSO is well suited for addressing optimization and NP-hard problems.

In the PSO framework, the optimization task is represented by an *objective function,* F_i. The goal is to find the maximum or minimum value of F_i, with this discussion focusing on maximizing F_i. The potential solutions of the optimization problem are called *particles*, the solution space is called *search space*, and the group of all particles is called *population*. Each particle has two attributes: *position* and *velocity*. The position of particle-i ($i \in \{1, \ldots, N_p\}$) in the search space is called *particle position* X_i, and the direction and speed of particle-i moving in the search space is called *particle velocity* V_i. First, the PSO algorithm initializes the position and velocity of each particle [29]. Then, we update the particle's position in each iteration [30]. We use the objective function F_i to evaluate the quality of particle-i's position. This process continues until a specific limit (e.g., the maximum number of iterations) is met. In each iteration, each particle needs to track two values to update its position.

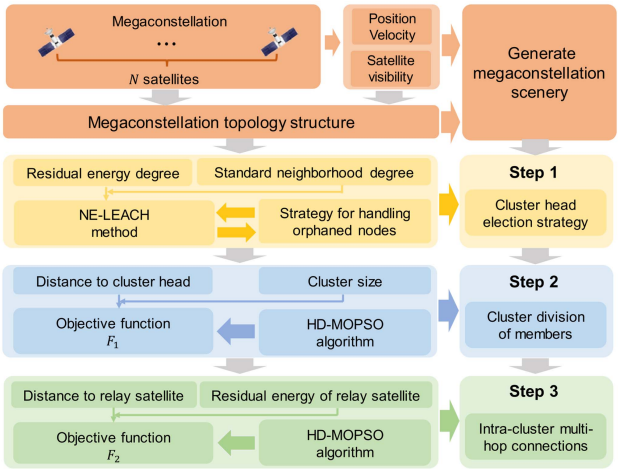


Fig. 4. Framework of megaconstellation ESC strategy, which includes three steps: cluster head selection strategy, member division into clusters, and intracluster multihop connections.

1) The individual optimal solution searched by each particle, i.e., the position X_{pi}. The individual optimal objective function value of the particle-i is F_{pi}.
2) The global optimal solution searched by the population, i.e., the position X_g. The population's optimal objective function value is F_g.

The position and velocity of particle-i are updated by

$$X_i(k) = X_i(k-1) + V_i(k-1) \tag{7}$$

$$V_i(k) = wV_i(k-1) + c_1a_1(X_{pi} - X_i(k-1)) + c_2a_2(X_g - X_i(k-1)) \tag{8}$$

where $k \in \mathbb{N}_+$ is the current iteration, $V_i(k)$ and $X_i(k)$ are the velocity and position at k iterations, $w > 0$ is the inertia weight, $a_1 \in [0, 1]$ and $a_2 \in [0, 1]$ are uniform random numbers utilized to increase the searching randomness, and $c_1 > 0$ and $c_2 > 0$ are the individual and global learning factors that regulate particles to their individual and global optimization.

However, the PSO algorithm is prone to slip into local optimum, resulting in poor optimization results. To avoid the local optimum, Hou et al. [31] proposed an HD-MOPSO algorithm, bringing a certain degree of mutation. We increase the population's uncertainty by adding random variation to certain particles. We select particles with a certain probability for initialization. The HD-MOPSO algorithm uses a local search strategy to accelerate the algorithm convergence. Denote a threshold

$$F_{\text{ave}}(k) = \frac{1}{N_p}\sum_{i=1}^{N_p} F_i. \tag{9}$$

In the local search phase, we evaluate whether $F_i < F_{\text{ave}}(k)$. If this condition is met, the particle-i is considered in a poor

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Algorithm 1: HD-MOPSO Algorithm.

Input : Maximum iterations I_{max}, Mutation probability P_m, Total particles N_p, Objective function F_i

Output: Global optimal solution X_g

Initialize $X_i(k), V_i(k)$;
for $k = 1$ **to** I_{max} **do**
  for *particle* $i = 1$ **to** N_p **do**
    Update X_i ,V_i (using Eqs. (7) and (8))
    if $F_i > F_{pi}$ **then**
      $X_{pi} = X_i$;
    if $F_i > F_g$ **then**
      $X_g = X_i$;
  for *particle* $i = 1$ **to** N_p **do**
    if *Rand[0,1]* $< P_m$ **then**
      $X_i(k) =$ Random position in search space;
  for *particle* $i = 1$ **to** N_p **do**
    if $F_i < F_{ave}(k)$ **then**
      Update $X_i(k)$ (using Eqs. (9) and (10));
return X_g

position. Then, we update the position $X_i(k)$ using

$$X_i(k)|_{\text{new}} = \frac{1}{2}(X_i(k)|_{\text{old}} + X_g) \tag{10}$$

where $X_i(k)|_{\text{new}}$ is the updated position, and $X_i(k)|_{\text{old}}$ is the old position. Algorithm 1 presents the workflow of the HD-MOPSO algorithm.

III. MEGACONSTELLATION CLUSTERING STRATEGY

To address the constellation clustering problem, we propose the ESC strategy, whose overall flowchart is shown in Fig. 4, consisting of the following three steps.

Step 1. Cluster Head Election Strategy: Based on the constellation topology, we strategically select a series of cluster heads using the NE-LEACH strategy with an orphan handing mechanism to provide full coverage of the constellations.

Step 2. Cluster Division of Members: We formulated a multiobjective function considering the distance of members to the cluster head and the cluster size. We applied the HD-MOPSO method to optimize this objective function for effective division.

Step 3. Intracluster Multihop Connections: We design a multiobjective function that incorporates both the satellite-to-relay distance and the relay satellite's residual energy reserves. This problem is solved using the optimization approach as well.

A. Cluster Head Satellite Election Strategy

The ESC strategy's cluster head election includes the NE-LEACH method and a process for handling orphaned nodes.

1) *NE-LEACH Method:* The LEACH method is widely used for cluster head election in wireless sensor networks [14]. The algorithm's execution time is divided into equal time-constant rounds, each with a duration of T. At the beginning of each round, each satellite node generates a random number $\mathcal{R}_i \in [0, 1]$, $i \in \{1, 2, \ldots, N\}$. A node will be selected as the cluster head if $\mathcal{R}_i < \mathbb{T}(n_i)$ with

$$\mathbb{T}(n_i) = \begin{cases} \dfrac{p}{1 - p\left(r \bmod (1/p)\right)}, & \text{if } n_i \in \mathbb{G} \\ 0, & \text{otherwise} \end{cases} \tag{11}$$

where $p > 0$ is the proportion of satellite node designated as cluster head, and r is the current number of rounds. Each satellite can only be elected as cluster head once in each $1/p$ round. $\mathbb{G}$ represents the set of satellite nodes that were not chosen as cluster heads in the previous $1/p$ round, i.e., the cluster head satellite candidate set. LEACH's random cluster head selection can lead to low-energy satellites being chosen, increasing their energy consumption and mortality rate. To enhance clustering, we present the NE-LEACH method, which builds upon LEACH by incorporating satellite residual energy and neighborhood degree. To quantify the ith satellite's residual energy E_i, we define the residual energy degree

$$\lambda_i = \frac{E_i}{E_{\max}}. \tag{12}$$

The neighborhood degree γ_i quantifies the count of the ith node's connections within the current topology, i.e., sum of elements in the ith row of adjacency matrix A, $\gamma_i = \sum_{j=1}^{N} A_{ij}$. Then, we normalize γ_i to get a standard version

$$\gamma_{si} = \frac{\gamma_i}{\gamma_{\max}} = \frac{1}{\gamma_{\max}} \sum_{j=1}^{N} A_{ij} \tag{13}$$

where $\gamma_{\max} = \max\{\gamma_i|_{i \in n_i}\}$ in the current round. Thus, we propose an enhanced version of (11) as follows:

$$\mathbb{T}(n_i) = \begin{cases} \dfrac{p(\varphi\lambda_i + (1-\varphi)\gamma_{si})}{1 - p\left(r \bmod \left(1/p\right)\right)}, & \text{if } n_i \in \mathbb{G} \\ 0, & \text{otherwise} \end{cases} \tag{14}$$

where $\varphi > 0$ is the weight.

2) *Strategy for Handling Orphaned Nodes:* NE-LEACH selects $p \times N$ initial cluster heads from satellite nodes at the start of each round, with the remaining nodes designated as members. However, Earth's physical masking can prevent some members from establishing communication links with any cluster heads. We refer to such a member as an *orphan node s*. Denote the orphan node set $\mathcal{S} = \{s_1, s_2, \ldots, s_{N_o}\}$ with a total number N_o. We further present a three-step orphan node handling strategy to address this issue.

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Step 1: Reuse the NE-LEACH method for $\mathcal{S}$. Select a new cluster head from $\mathcal{S}$.

Step 2: Remove the selected head from $\mathcal{S}$. Update $\mathcal{S}$ and check if $s_i \in \mathcal{S}$.

Step 3: Repeat Steps 1 and 2 if $\mathcal{S} \neq \varnothing$, until $\mathcal{S} = \varnothing$.

REMARK 2 In the NE-LEACH mechanism, if a node is selected as cluster head in the $1/p$ round, it is removed from the cluster head candidate set $\mathbb{G}$. However, when using the NE-LEACH method on $\mathcal{S}$, there may be some nodes in $\mathcal{S}$ that were previously elected as cluster heads. Hence, not all orphan nodes are in $\mathbb{G}$. This may result in these nodes not being selected as cluster heads. Thus, in the orphan handling strategy, we use $\mathcal{S}$ as a candidate set for the cluster head instead of the satellites in $\mathbb{G}$.

REMARK 3 The NE-LEACH mechanism's cluster head election is influenced by two main factors: 1) weight φ. Adjusting φ affects the selection tendency: increasing φ makes satellites with higher residual energy more likely to be chosen as cluster heads, while decreasing φ favors satellites with higher neighborhood degrees. 2) Proportionality factor of cluster heads to total number of nodes p. p depends on the need for clusters in the constellation design process. In addition, the constellation-building process is step-by-step. We can adapt p according to the needs of constellation management.

B. Cluster Division of Member Satellites

We further optimize the cluster division of members. To solve this problem, we formulate the optimization objective using an energy- and cluster-size-based function.

1) *Energy Consumption:* Equation (6) highlights that distance significantly influences energy consumption. Hence, we can reframe the energy consumption optimization problem as a distance-based optimization challenge. Our objective is to divide members into the nearest clusters.
2) *Cluster Size:* The cluster head oversees its members, and an excessive number of members place a substantial management burden on the head, resulting in additional energy expenditure. Consequently, we aim for equitable distribution of members across each cluster.

The objective function for the ith member $\mathcal{M}_i$ is given by

$$f_{1i}(D_{ij}, N_M^i) = \sigma_1 \frac{D_{ij}}{D_{\max}} + (1-\sigma_1)\left|\frac{N_M^i - N_{\text{aver}}}{N_{\text{aver}}}\right| \quad (15)$$

where $i \in \{1, \ldots, m\}$ denotes the number of members. Due to intersatellite communication constraints, the number of cluster heads that each member satellite can connect to differs. We define $\mathcal{H} = \{H_1, \ldots, H_m\}$, where H_i denotes the number of cluster heads that member $\mathcal{M}_i$ can construct for communication connections. Given an $\mathcal{M}_i$ with $j \in \{1, \ldots, H_i\}$, we can change $f_{1i}(D_{ij}, N_M^i)$ by adjusting the connections between member satellites and different cluster heads, where D_{ij} is the distance between member $\mathcal{M}_i$ and head C_j; N_M^i is the size of cluster $\mathcal{C}_j$ to which the member $\mathcal{M}_i$ is linked; $N_{\text{aver}} = m/H$ is the cluster size based on the average distribution of members; and $\sigma_1 > 0$ is the weight.

The objective function of total members $\mathcal{M}$ is given by

$$F_1 = \sum_{i=1}^{m} f_{1i}(D_{ij}, N_M^i). \quad (16)$$

Hence, we transform the member cluster division problem into $\min\{F_1\}$. Then, we optimize F_1 for member $\mathcal{M}_i$ using the HD-MOPSO method. We define all members $\mathcal{M}_i$ as a particle. The set of cluster heads C to which members can be connected is defined as the search space, and these heads are chosen to satisfy the requirements of the communication links with the members. We minimize F_1 by adjusting the connections between members and different cluster heads. In addition, there is a tradeoff between the two objectives in F_1. Members may choose to join in larger clusters to reduce energy consumption.

REMARK 4 In practical applications, σ_1 can be adjusted to balance the two objectives in F_1 by prioritizing one over the other. Given the limited battery capacity of satellites, communication energy consumption, primarily determined by intersatellite distance, is a key factor. In addition, reducing the distance between members and cluster heads enhances communication efficiency. Therefore, in the optimization process, increasing σ_1 raises the priority of energy consumption in the objective function.

C. Intracluster Multihop Connectivity

We present an intracluster multihop connection approach to achieve energy-saving purposes further. As shown in (6), the energy consumption model mainly depends on D^2. Due to the nonlinear nature of D^2, the longer the transmission distance, the energy consumption increases in square steps. Therefore, long-distance transmission can be divided into multiple short-distance transmissions by relaying to reduce energy consumption further. As shown in Fig. 2, in $\mathcal{C}_1$, C_1^1 as a source satellite, far from C_1, can connect with C_1 via C_1^2 as a relay satellite. C_1^2, as a relay satellite, receives *hello* message from C_1^1 and obtains the geographic position and ID information of C_1^1. Subsequently, C_1^2 forwards *hello* message to C_1, conveying the geographic position and ID information of C_1^1. Similarly, member C_1^3 needs C_1^2 as a relay satellite when obtaining information from head C_1. According to (6), as a relay satellite, C_1^2 consume more energy.

The intracluster multihop connectivity problem is similar to the intracluster membership division problem. Therefore, to solve this problem, we similarly construct a suitable objective function. We consider the following two objectives.

1) *Relay Satellite's Residual Energy:* Source satellites should be closer to their relay satellites to minimize

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energy consumption, especially by shortening their distance.

2) *Distance to Relay Satellite:* According to (6), the relay satellite consumes more energy. Therefore, we hope the relay satellite to have more residual energy.

For a given cluster $\mathcal{C}_j$, there are N_{L_j} source satellites that require multihop connections. Therefore, for $\mathcal{C}_j$, the objective function of the ith source satellite is

$$f_{2i}^{j}(D_{i\mu}^{j}, E_{\mu}^{j}) = \sigma_2 \frac{D_{i\mu}^{j}}{\mathbb{D}_{\max}^{j}} + (1-\sigma_2)\frac{\mathbb{E}_{\max}^{j}}{E_{\mu}^{j}} \tag{17}$$

where $j \in \{1, \ldots, H\}$ is the number of clusters; $\mathbb{D}_{\max}^{j}$ denotes the maximum distance between all source satellites and all relay satellites within the $\mathcal{C}_j$; $\mathbb{E}_{\max}^{j}$ denotes the maximum energy of all relay satellites in $\mathcal{C}_j$; $i \in \{1, \ldots, N_{L_j}\}$ is the number of source satellites in $\mathcal{C}_j$; and $\sigma_2 > 0$ is the weight. In $\mathcal{C}_j$, not each source satellite can establish communication links with all relay satellites. Hence, in $\mathcal{C}_j$, we define set $\mathbb{L}_j = \{L(1), \ldots, L(N_{L_j})\}$, where $L(i)$ denotes the number of relay satellites that can construct a communication link with the ith source satellite. $D_{i\mu}^{j}$ is the distance between source satellite i and relay satellite μ. E_{μ}^{j} is the residual energy of μ.

Thus, the objective function for all source satellites is

$$F_2 = \sum_{j=1}^{H} \sum_{i=1}^{N_{L_j}} f_{2i}^{j}(D_{i\mu}^{j}, E_{\mu}^{j}). \tag{18}$$

We transform the intracluster multihop connectivity problem to $\min\{F_2\}$. Specifically, we define all source satellites as a particle. We define relay satellites as search space, and they can build communication links with source satellites. Then, we can use the HD-MOPSO algorithm to optimize F_2 similarly as we optimize F_1.

REMARK 5 The optimization function F_2 takes the shortest distance between satellites and the maximum residual energy of the relay satellite as the objective of selecting the relay satellite. However, there is a tradeoff between these two objectives, and the relay satellite with the shortest distance is not necessarily the one with the most energy. Considering that energy consumption is a key factor and is related to distance, we balance these two objectives by increasing the weight σ_2 to optimize the performance of the objective function.

REMARK 6 In solving the intracluster multihop connection optimization problem, the source satellite evaluates the objective function F_2 by selecting different relay satellites. For small clusters, the optimal value of F_2 can be determined by evaluating the connections between the source satellite and each relay satellite individually. However, as the cluster size increases, this method becomes computationally expensive, slowing down the solution process. To improve efficiency, optimization methods can be employed in larger clusters to speed up the solving process and avoid extensive calculations.

TABLE I
Parameter Settings in Simulation

Items	Value
Duration of a single periods T	60 s
Running time rounds	120
Maximum communication distance D_{max}	500000 m
Cluster head percentage p	0.03
Number of satellites in megaconstellation N	636
Number of multi-hop in the cluster $\mathcal{K}$	2
Energy consumption for a 1-bit transmission E_{ele}	1×10^{-8}J
Radio-frequency amplifier gain ε_{amp}	5×10^{-12}J
Weighting factors in $\mathbb{T}(n_i)$ φ	0.2
Weighting factors in F_1 σ_1	0.7
Weighting factors in F_2 σ_2	0.8

REMARK 7 For megaconstellations, such as Starlink, communication between terminals usually requires multiple hops. We have designed an intracluster multihop transmission method that requires only two hops. Thus, the reliability of the communication is sufficient. Through the intracluster multihop transmission method, we can reduce the communication distance between satellites and improve communication efficiency. According to (6), we further optimize the energy consumption through multihop transmission.

REMARK 8 This article formulates a two-step optimization problem to address the cluster division of members and intracluster multihop communication, employing the HD-MOPSO algorithm to solve these problems. Compared to the traditional PSO algorithm, the HD-MOPSO algorithm incorporates a local search strategy, enhancing the convergence speed of the solution. In addition, the following two strategies are proposed to further accelerate the optimization process. 1) *Optimization starting point design:* the division method that assigns members to the nearest cluster head is used as an initial solution for one of the particles. This approach improves the quality of the algorithm's initial solutions. 2) *Optimization threshold introduction:* the iteration process stops early if the change in the globally optimal value remains below a predefined threshold for consecutive iterations.

IV. SIMULATION FOR MEGACONSTELLATION

This section evaluates the proposed ESC strategy, examining the performance, cluster count, size, and energy consumption.

A. Simulation Settings

Simulations were conducted to evaluate the performance of the OneWeb system, consisting of 636 satellites distributed across 18 circular orbits at an altitude of 1200 km and an inclination of 87°. In the simulation, data are collected from the constellation scenario every round, with a round duration of $T = 60$ s. The total simulation time is 2 h, consisting of 120 rounds, with each round lasting T. Table I

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gives the strategy parameter settings. The following three metrics are used to evaluate strategy performance.

1) *Number of Clusters:* To reduce the burden of managing megaconstellation at ground stations, we hope to reduce the number of clusters created.
2) *Cluster Size:* To evaluate the size of clusters, we use

$$F_{\text{cs}} = \sum_{i=1}^{N} \left| \frac{N_s^i - N_{\text{aver}}}{N_{\text{aver}}} \right| \tag{19}$$

where $i \in \{1, \ldots, N\}$ denotes the number of satellites in the constellation, and N_s^i denotes the size of the cluster to which the ith satellite belongs. A smaller F_{cs} indicates a more balanced cluster and fewer large clusters.
3) *Energy Consumption:* Since the satellite has a limited battery capacity, we hope to reduce its energy consumption and extend its life in orbit.

REMARK 9 According to (20), the load balancing factor $\mathcal{L}$ is equally sensitive to larger and smaller clusters. However, in practical scenarios, a larger cluster increases the management pressure on the head. Therefore, the impact of larger cluster size and smaller cluster size on the load balancing factor should not be the same. In (19), we quantify the uniformity of cluster size by F_{cs}. Larger clusters have a more significant effect on F_{cs}, while smaller clusters have less effect. Hence, F_{cs} can reflect the balance and size of clusters.

B. Simulation Scheme Design

1) *Number of Clusters:* The WCA method, a traditional approach for cluster head selection in wireless sensor networks [18], [32], calculates a weight function based on neighborhood degree, mean distance to neighbors, and residual energy [32]. This function guides cluster head selection for each node. This section compares the number of clusters generated by WCA, LEACH, and the proposed cluster head election strategy and cluster strategy in [13] when applied to OneWeb constellations.

REMARK 10 Setting p to 0.03, based on LEACH experience, both NE-LEACH and LEACH select $p \times N$ cluster heads using their respective mechanisms. Similarly, WCA and cluster strategies in [13] appoint $p \times N$ cluster heads. If these initial selections do not achieve full constellation coverage, resulting in orphan nodes, additional cluster heads are elected using the respective method until complete coverage is achieved.

2) *Cluster Size:* The cluster size is primarily influenced by how its members are divided. When designing experiments to compare cluster sizes, we focus on the issue of member division. In Section III-B, we transform the member cluster division problem into solving $\min\{F_1\}$. We optimize F_1 by comparing the discrete gradient optimization, minimum distance, PSO, and HD-MOPSO approaches. The cluster size is quantified using F_{cs} as defined in (19). In addition, we compare the cluster strategy in [13].

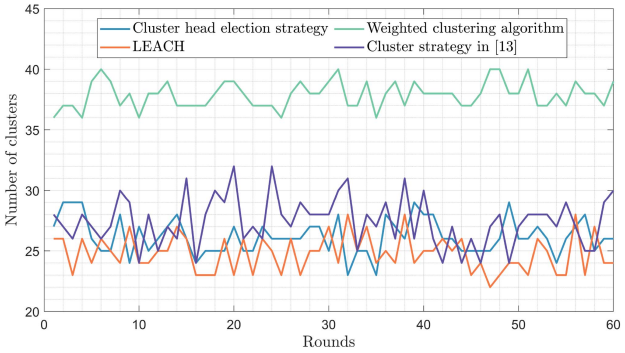


Fig. 5. Number of clusters generated per round by LEACH, WCA, cluster head election strategy, and cluster strategy in [13], where the average number of clusters generated by each of the four methods under 60 rounds is 24.8, 37.8, 26.1, and 27.5, respectively.

3) *Energy Consumption:* In designing the energy consumption experiment, we consider the following two cases. 1) Members and cluster heads are connected via a single hop. We only need to analyze the member cluster division issue. We optimize F_1 by discrete gradient optimization, minimum distance, PSO, and HD-MOPSO, respectively. 2) Members are connected to the cluster head via single-hop or multihop. In Sections III-B and III-C, we provide a detailed description of cluster division of members and intracluster multihop. Here, we use HD-MOPSO to optimize F_1 and F_2 (called HD-MOPSO+multihop). In addition, we use $\mathcal{K} = 2$ hops for multihop in experiments. Then, we design energy consumption experiments using discrete gradient optimization, minimum distance, PSO, HD-MOPSO, and HD-MOPSO+multihop, respectively. Again, we use the cluster strategy in [13] as a comparison for energy consumption.

C. Performance Analysis

1) *Number of Clusters Analysis:* Fig. 5 presents a comparative analysis of the cluster quantities generated by WCA, LEACH, the cluster head election strategy, and the cluster strategy in [13]. In successive rounds, the number of clusters for each method fluctuated within a range, and the number of clusters remained relatively stable. For clarity, we selected data from the first 60 rounds.

Over 60 rounds, WCA generated an average of 37.8 clusters, approximately $6\% \times N$ clusters, compared to 24.8, 26.1, and 27.5 clusters (approximately $4\% \times N$ clusters) produced by the other three methods. Compared to other methods, WCA generates a higher number of clusters, exacerbating the complexity of ground network management for the constellation. WCA methods face significant disadvantages in cluster generation. Due to the uneven distribution of satellite density in the constellation, satellites in certain regions (e.g., high-latitude areas) may exhibit superior characteristics, such as speed, neighborhood degree, and intersatellite distance. However, the WCA relies on generic deterministic criteria for cluster head selection, which can lead to cluster heads being concentrated in specific regions (e.g., high-latitude areas). The Earth's obstruction

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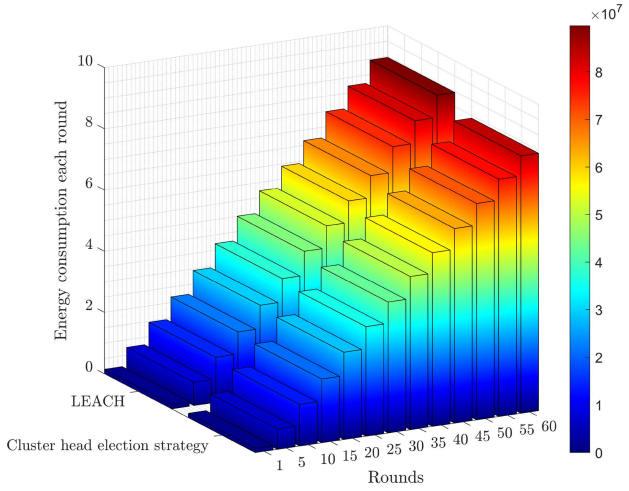


Fig. 6. Comparison of energy consumption between LEACH and the proposed cluster head election strategy.

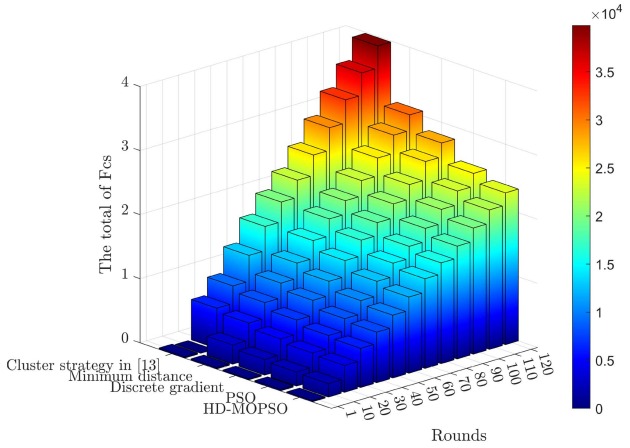


Fig. 7. Comparison of F_{cs} when different methods are used, including optimization of F_1 using the discrete gradient method, the minimum distance, PSO, HD-MOPSO, and adoption of clustering strategy in [13].

makes communication between different regions challenging, necessitating more cluster heads to cover the entire constellation.

For example, in the OneWeb constellation, satellite density is much higher in high-latitude regions than in low-latitude regions, leading to better neighborhood degree, relative speed, and intersatellite distance. When WCA is used for cluster head selection, cluster heads may predominantly appear in high-latitude regions, making it difficult for them to cover satellites in low-latitude regions directly [33]. Therefore, additional cluster heads are required to ensure full constellation coverage. However, LEACH and the proposed cluster head election strategies by employing $\mathbb{T}(n_i)$ for selection, introduce randomness, resulting in a more uniform distribution of cluster heads across the constellation. Similarly, the cluster head selection strategy in [13] incorporates random values while accounting for the nodes' residual energy and the Euclidean distance between them. This approach reduces the number of cluster heads required to cover the constellation. We further discuss the clustering strategy designed in [13] in terms of cluster size and energy consumption metrics.

Since the proposed cluster head election strategy generates a similar number of clusters as the LEACH method, we further design energy consumption experiments to analyze the energy used by both methods. Specifically, cluster heads were selected in the OneWeb constellation using both methods, and members were assigned to clusters based on the minimum distance to the cluster head. As shown in Fig. 6, the proposed strategy consumes 6.3% less energy than LEACH over 60 rounds. Compared to the LEACH method, the NE-LEACH method in the proposed cluster head selection strategy incorporates neighborhood degree and residual energy factors, leading to improved energy efficiency. Compared to WCA and LEACH, the proposed strategy effectively identifies suitable cluster heads while reducing cluster count, providing a strong foundation for member cluster division.

2) *Cluster Size Analysis:* In large-scale networks, the load-balancing factor $\mathcal{L}$ is commonly used to quantify the degree of cluster equalization [26]

$$\mathcal{L} = \frac{H}{\sum_{i=1}^{H}(N_c^i - N_{\text{aver}})^2} \tag{20}$$

where N_c^i is the size of cluster $\mathcal{C}_i$, and $i \in \{1, \ldots, H\}$ is the number of cluster.

Large cluster sizes place a heavy load on the corresponding cluster head. While (20) suggests a similar influence on $\mathcal{L}$ for both large and small clusters, we introduce F_{cs} to evaluate clustering balance. As defined in (19), F_{cs} is more sensitive to larger clusters, reflecting their greater impact on balance. This approach considers both cluster balance and size. We optimize F_1 using discrete gradient optimization, minimum distance, PSO, and HD-MOPSO to conduct member cluster division. We compare the cluster strategy in [13].

Fig. 7 shows that the HD-MOPSO algorithm is better than discrete gradient optimization, PSO, minimum distance methods, and cluster strategy in [13]. The minimum distance method assigns members to the nearest cluster, optimizing energy consumption but overlooking the critical factor of cluster size. The discrete gradient optimization method iteratively adjusts parameters along the negative gradient to find the optimal F_1, while PSO achieves the same goal by iteratively updating particle positions. The objective function F_1 depends on two key metrics: the distance between members and heads, and cluster size. While the distance metric is easy to determine, the cluster size metric is influenced by how members are assigned to clusters. In addition, there is a tradeoff between the objectives within F_1 [as shown in (16)]. These factors lead discrete gradient optimization and PSO to local optima, resulting in suboptimal outcomes. Compared to discrete gradient optimization, PSO leverages its random search capability, which enhances its optimization performance.

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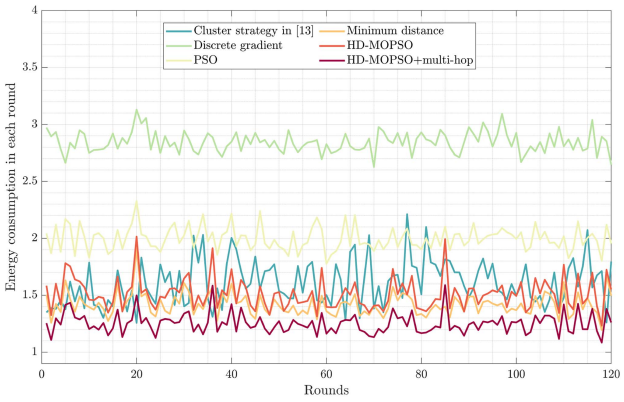


Fig. 8. Discrete gradient optimization, PSO, minimum distance, HD-MOPSO, HD-MOPSO+multihop, and cluster strategy in [13] energy consumption for each round.

TABLE II
Energy Consumption Comparison Across Different Methods

Algorithm	Energy consumption	Comparison
Discrete gradient optimization	$3.412 * 10^8$J	Baseline
PSO	$2.397 * 10^8$J	↓ (29.7%)
Cluster strategy in [13]	$1.93 * 10^8$J	↓ (43.4%)
Minimum distance	$1.703 * 10^8$J	↓ (50.1%)
HD-MOPSO	$1.8127 * 10^8$J	↓ (46.9%)
HD-MOPSO+multi-hop	$1.5047 * 10^8$J	↓ (55.9%)

In contrast, HD-MOPSO incorporates local search and mutation mechanisms, effectively avoiding these local optima. As illustrated in Fig. 7, HD-MOPSO achieves superior results.

3) *Energy Consumption Analysis:* Fig. 8 shows the energy consumption per round. Table II benchmarks this energy consumption against several algorithms using discrete gradient optimization as a baseline.

The tradeoff between the two objectives in F_1 leads to the creation of local optima. Discrete gradient optimization can easily fall into these local optima and is not optimized well. In addition, PSO has a strong global search capability, but its stochastic particle motion leads to energy consumption fluctuations under different rounds. Both methods are prone to local optimization, leading to excessive energy consumption. The clustering strategy proposed in [13], which considers the intersatellite distance metric when assigning members to clusters, achieves a 43.4% reduction in energy consumption compared to the discrete gradient optimization method. However, using the HD-MOPSO algorithm efficiently avoids this problem. Due to cluster size constraints in F_1, HD-MOPSO consumes more energy than the minimum distance method. We further optimize the energy consumption by using the intracluster multihop connections method. Fig. 8 shows that we consume less energy via HD-MOPSO+multihop than the minimum distance approach. As given in Table II, using discrete gradient optimization as the baseline, the PSO, minimum distance, HD-MOPSO, and HD-MOPSO+multihop strategies achieved energy savings of 29.7%, 50.1%, 46.9%, and 55.9%, respectively.

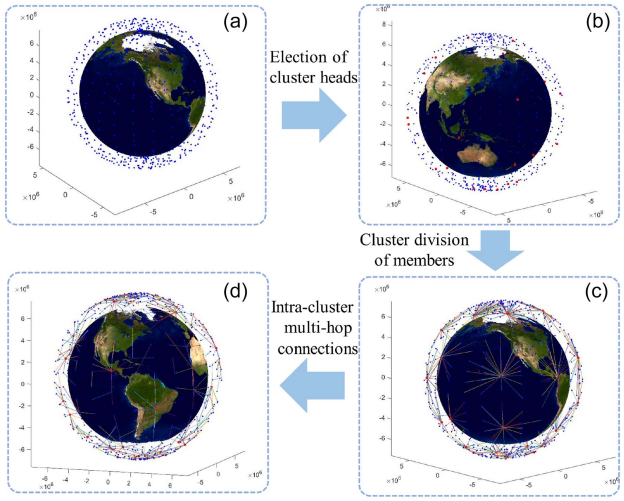


Fig. 9. Cluster division process in the OneWeb constellation is illustrated as follows: (a) depicts the initial node distribution within the constellation, (b) shows the node distribution after selecting cluster heads (red nodes represent cluster heads and blue nodes represent members), (c) demonstrates the topological connections between satellites after assigning members to clusters, and (d) highlights the satellite topology following the establishment of intracluster multihop connections.

D. Simulation Summary

Fig. 9 illustrates the OneWeb constellation clustering process and results using the ESC strategy. Fig. 9(a) depicts the initial spatial distribution of constellations without clustering. Fig. 9(b) shows the distribution of selected cluster heads. Fig. 9(c) presents the cluster divisions after applying orphan nodes coping strategies. Finally, Fig. 9(d) incorporates intracluster multihop connectivity. The results demonstrate successful cluster head election or cluster joining for all satellites, with some utilizing relay communication within their clusters.

V. CONCLUSION

In this article, we propose the ESC strategy to efficiently divide constellations into clusters. The ESC strategy utilizes a cluster head election method combined with an orphan handling strategy to select appropriate cluster heads. During the selection process, the NE-LEACH method is integrated, considering both satellite neighborhood degree and residual energy. Following this, a multiobjective optimization problem is designed for cluster division of members and intracluster multihop connections, respectively, and solved using the HD-MOPSO algorithm. Moreover, the tradeoff relationship between the objectives is considered when solving the optimization problem. Simulation results for the OneWeb constellation demonstrate the superior performance of the ESC strategy. The proposed cluster head selection method achieves full constellation coverage with fewer cluster heads than the WCA method. Moreover, ESC achieves a more balanced cluster size and energy consumption reduction compared to discrete gradient optimization. Future work will focus on integrating megaconstellation

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management with space–ground networks and developing hierarchical strategies across multiple orbital altitudes.

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